Mohit Choudhary

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.68
MAHESHWARI PUBLIC SCHOOL, PRATAP NAGAR, JAIPUR	Class XII (CBSE)	2023	89.20%
MAHESHWARI PUBLIC SCHOOL, PRATAP NAGAR, JAIPUR	Class X (CBSE)	2021	97.20%

PROJECTS

Formula Student EV & ICV Development

- Heading the Chassis domain and leading the **design and fabrication** of the **chassis** for the **Formula Bharat 2027** Electric Vehicle, developing structural geometry and weldment assemblies on **SOLIDWORKS** and validating through **Static Structural Analysis** on **ANSYS**.
- Performing **load analysis** and **FEA-based design evaluations** to optimize component geometry and ensure structural integrity; conducted frontal impact simulations on ANSYS to evaluate the **Factor of Safety (FOS)** of the chassis under crash loading conditions, ensuring compliance with Formula Student safety regulations.
- Represented IIT Hyderabad at the **12th SUPRA SAEIndia 2025** Formula Student competition at Buddh International Circuit, Greater Noida, contributing to the manufacturing phase of the ICV formula student car.

House Price Prediction Model

- Developed a machine learning model to predict residential property prices using features such as location, area, bedrooms, and bathrooms by performing data cleaning, preprocessing, and feature engineering.
- Evaluated **Linear Regression, Lasso Regression, and Decision Tree Regressor** models, comparing their performance to identify the most accurate predictor using appropriate evaluation metrics.

Netflix Data Analysis using Python

- Performed exploratory data analysis on **8,000+ Netflix titles**, cleaning and preprocessing the dataset using Pandas.
- Created **15+ visualizations** using Matplotlib and Seaborn to analyze content trends, genres, ratings, release patterns, and country-wise distributions.
- Derived actionable insights through statistical analysis and visualization to identify content trends and audience preferences.

GIS-Based Assessment of Rainfall Variability

- Conducted a remote sensing study of Sangareddy District (Hyderabad) using **Google Earth Engine (GEE)**, **analysing rainfall trends over a 10-year** period (2016-2025) using the CHIRPS daily rainfall dataset.
- Applied **NDWI (Normalized Difference Water Index)** on Sentinel-2 satellite imagery to map and quantify surface water bodies, recording a **water area reduction** from 17 million m^2 (2022) to 16 million m^2 (2023).
- Compared reservoir and surface water conditions across years to identify variability patterns, demonstrating the applicability of **GIS and remote sensing** tools for regional water resource monitoring.

SKILLS

Programming: C, C++, Python, SQL, Data Structures & Algorithms (DSA) | **Data & ML Libraries:** NumPy, Pandas, Scikit learn, Matplotlib, Jupyter Notebook | **CAD & Simulation:** SOLIDWORKS, ANSYS, Google Earth Engine (GEE) | **Soft Skills:** Leadership, Team Management, Critical Thinking, Problem Solving

POSITIONS OF RESPONSIBILITY

Sponsorship Coordinator, IISM Organizing Team, **IIT Hyderabad**

- **Coordinated cross-functional sponsorship execution** for the **58th Inter-IIT Sports Meet 2025**, managing external

stakeholder communication (Red Bull), MoU coordination, and delivery tracking across multiple teams for a large-scale event with **3,800+ athletes**.

Internship Coordinator, **Office of Career Services, IIT Hyderabad**

- **Managed internship hiring workflows** by maintaining recruiter pipelines, coordinating recruitment drives, tracking student applications, and streamlining documentation across multiple stakeholders.

Core, Ideation Domain, **Tinkerers' Lab, IIT Hyderabad**

- **Planned and executed large-scale technical events**, including **THRUST 2026** and the **AVEVA National AI/ML Hackathon**, coordinating cross-functional teams to manage **6,000+ registrations** from planning through final presentations.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications

Mathematics & Statistics: Calculus, Linear Algebra, Differential Equations, Transform Techniques, Probability & Random Variables, Introduction to Statistics

Core engineering: Mechanics of Solids, Structural Analysis, Fluid Mechanics, Thermodynamics, Geotechnical Engineering, Digital Fabrication, Engineering Mechanics

EXTRACURRICULAR

- Awarded 2nd place in Astrophotography Competition at Milan (inter-hostel competition) for excellence in technical and creative photography.
- Secured 3rd Place in the Bridge Making Competition organized at the Civil Engineering Department Day, IIT Hyderabad.